

A Leaf-shaped Antipodal Vivaldi Antenna for Robust Wireless Communications in Industrial IoT Applications

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Abstract— This study proposed a leaf-shaped antipodal Vivaldi antenna (AVA) that operates in the 10 GHz to 35 GHz frequency band and was designed specifically for reliable wireless communication in industrial Internet of things applications. Using a Teflon substrate ($\epsilon_r = 2.1$, $\tan\delta = 0.0002$), the antenna was built and powered by a 50 Ω microstrip feedline. A leaf structure with two bending branches was integrated into the design, greatly improving the antenna's performance. Three distinct design techniques were employed to evaluate performance: a large leaf structure, a large width of bending, and a smaller leaf structure. The proposed antenna with its medium leaf structure and optimized bending width demonstrated superior performance, achieving a peak gain of 12.2 dB at 30 GHz and a radiation efficiency of 97.8% at 16.25 GHz.

Keywords—Antipodal Vivaldi antenna (AVA), wireless communication, microwave, mm-wave, leaf structure.

I. INTRODUCTION

The modern world relies on the Internet of Things (IoT) to make commonplace things smarter and more efficient by linking them to the internet. This technology enhances our lives by automating tasks and improving industrial processes with real-time data, such as monitoring and optimizing production lines in manufacturing plants [1]. Traditional antennas have several limitations that affect their performance in modern applications. They often have narrow bandwidths, limiting their ability to handle various frequencies. Additionally, they may provide insufficient gain and lower radiation efficiency, which can weaken signal strength and reduce overall effectiveness [2]. An antenna with a highly focused pattern and enhanced gain characteristics is needed to resolve these issues. Vivaldi Antenna is crucial for robust wireless communications in Industrial IoT (IoT) applications because it addresses the limitations of traditional antennas through its advanced design [3]. In industrial environments, where interference, physical obstructions, and harsh conditions can disrupt signals, this antenna provides high gain and ultra-wideband performance, ensuring reliable data transmission across a broad frequency range, including high-frequency bands essential for modern applications [4]. However, the Vivaldi antenna radiates horizontally, and it has end-fire radiation [5]. Vivaldi antennas come in three different varieties. Coplanar Vivaldi antennas (CVA), antipodal Vivaldi antennas (AVA), and balance antipodal Vivaldi antennas (BAVA) are the three types of such antennas. Generally, the Aperture feeding [6] or microstrip feeding technique [7] is used in coplanar Vivaldi antenna (CVA) and both flare shape is mounted on the same side of the substrate;

however, it has the drawback of limited bandwidth [8]. After that, the design and implementation of the BAVA were complex. The coplanar Vivaldi antenna's drawbacks are addressed by the Antipodal Vivaldi Antenna, which switches from an aperture coupling feed to a microstrip feed. This modification eliminates bandwidth limitations, with the flare shapes positioned on opposite sides of the substrate [9].

Numerous methods have been put forth in recent research to enhance the performance parameters of AVAs, including radiation pattern, efficiency, gain, and return loss. In [10], a fractal leaf structure and exponential tapering arm are used to design a small AVA where the leaf structure is more complex and the maximum gain is below 10 dB. Two different substrates FR4 and RO3006 are used to design a small AVA in [11], where the maximum gain is below 6 dB. AVA with an additional slot is proposed in [12] which yields high radiation gain with more than 90% radiation efficiency at the frequency range of 1-13 GHz; however, its return loss is not UWB like other conventional AVA. A large-size AVA with elliptical corrugated is proposed in [13] to improve gain and radiation pattern but the size increases the cost. Compact AVA with a bending feedline structure is proposed in [14] which maximum gain and efficiency are 5.2 dB and 83% respectively. AVA with an elliptical parasitic patch is proposed in [15] for improving radiation pattern and gain, however, it provides low gain at high frequency.

In this paper, the antenna has two radiating arms on either side of the substrate, each containing leaf structures. To achieve maximal radiation in the end-fire direction, the traveling wave current phases on each arm must be 180° apart. By adjusting the positioning of these arms and varying the size of the leaf, the efficiency, gain, directivity, return loss, and radiation characteristics are enhanced.

II. GEOMETRY OF THE PROPOSED ANTENNA

The suggested AVA antenna (AVA-P) structure and its dimensions are depicted in Figure 1. The substrate material is Teflon [polytetrafluoroethylene (PTFE)] with a thickness of 0.8 mm, characterized by a dielectric constant (ϵ_r) of 2.1 and a loss tangent ($\tan\delta$) of 0.0002, with a dimension of 49 mm $\times$ 54 mm ($W_{\text{sub}} \times L_{\text{sub}}$). The conductor material is copper annealed with a thickness of 0.035 mm. Which is shaped like a leaf. To excite the antenna element, an exponential tapered microstrip line is used. To achieve the antenna's antipodal shape, two 180-degree phase-shifted arms are flared and two tapering flares (arms) are added to either side of the substrate.

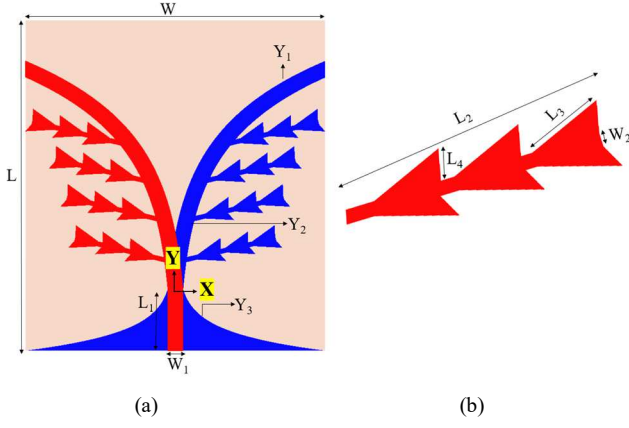


Fig. 1. (a) Transparent structure of the proposed AVA-P. (b) structure of the leaf of the AVA-P.

There are two exponential curves Y_1 and Y_2 , one polygon as a microstrip line feed ($L_1 \times w_1$), and two conducting plates (red flare acts as a radiating plane and blue flare acts as a ground plane) depicted in Fig. 1 (a). Both flares are mounted on the substrate surface. After that, the leaf structure was designed with both flares which is depicted in Fig. 1(b).

The geometrical parameters of the proposed AVA-P are shown in Figs. 1(a) and 1(b), and the optimal values for these parameters are given in Table I. In addition, the optimal exponential equation for the flare's curved profile is shown below.

$$y_1 = e^{pt} + a; \text{ where } 0 \leq t \leq 47 \quad (1)$$

$$y_2 = e^{qt} - b; \text{ where } 0 \leq t \leq 47 \quad (2)$$

$$y_3 = e^{-rt} - b; \text{ where } -10 \leq t \leq 0 \quad (3)$$

$$y_4 = e^{pt} + c; \text{ where } 0 \leq t \leq 37.5 \quad (4)$$

$$y_5 = e^{qt} - d; \text{ where } 0 \leq t \leq 34.41 \quad (5)$$

$$y_6 = e^{-st} - d; \text{ where } -10 \leq t \leq 0 \quad (6)$$

In this case, y_1 , y_2 , and y_3 indicate the flare's curved profile, whereas p , q , and r stand for the tapering rates used to manage the curved profile. The proper feedline width is chosen using the constant optimized values a , b , and c to provide impedance matching and effective power transfer between components. It is set to 2.56 mm for the feedline width to match the 50 Ω microstrip line.

TABLE I. OPTIMIZED VALUES OF THE PROPOSED ANTENNA'S GEOMETRICAL PARAMETERS

Parameter	Value (mm)	Parameter	Value (mm)	Parameter	Value (mm)
W	49	L_4	1.9	s	0.3184
W_1	2.56	L_5	5.21	q	0.8
W_2	0.8	L_6	1.9	a	2.28
L	54	L_7	3.5	b	0.28
L_1	10	p	0.088	c	2.8
L_2	15.5	q	0.092	d	0.8
L_3	4.7	r	0.3187		

The width of the opposite arm Y_1 and Y_2 and the size of the leaf L_3 and L_4 have a substantial influence on the gain, side lobe level, front-to-back ratio, and other factors in the design. Three configurations of the AVA, featuring a larger leaf shape of AVA (AVA-LL), increased width or arms of AVA (AVA-LW), and small leaf shape of AVA (SL) are illustrated in Fig.2. Table I shows the optimal values for these parameters.

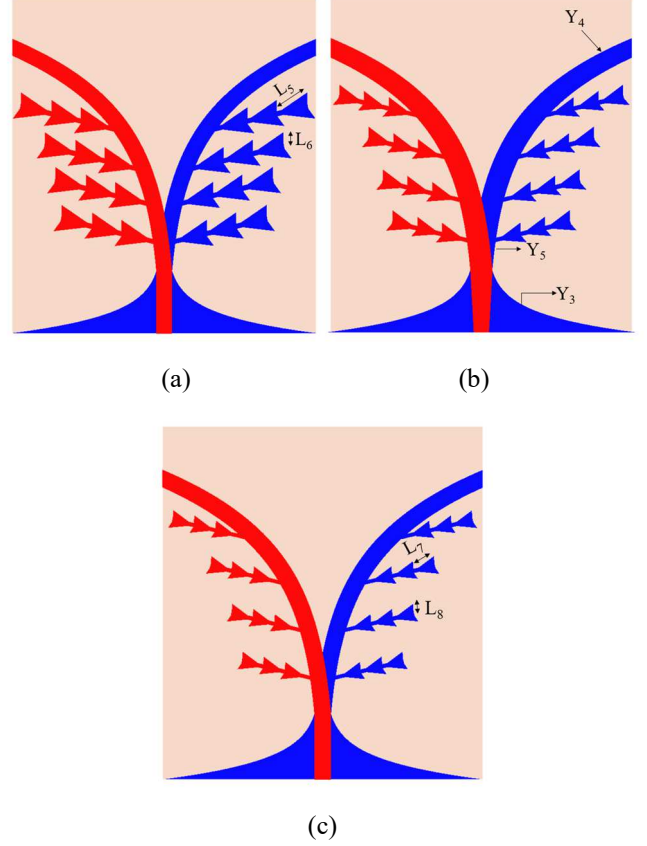


Fig. 2. Transparent view (a) structure of the proposed AVA with Larger leaf shape (AVA-LL). (b) the large width of bending AVA (AVA-LW). (c) small leaf of AVA (AVA-SL).

III. SIMULATION RESULTS

This section provided an in-depth analysis of the simulated results and evaluations of the AVA configurations, which incorporated various directors along with larger leaf shape AVA, large width of bending, and small leaf shape. Parameters like the radiation pattern, gain, efficiency, surface current distribution, and reflection coefficient were all analyzed and contrasted.

A. Reflection coefficient

Fig. 3 compares the simulated reflection coefficients for the AVA antennas, which are appropriate for Internet of Things applications and span the frequency range of 10 GHz to 35 GHz. Interestingly, all three patterns show values that are less than -10 dB. For each pattern, the maximum simulated reflection coefficient values are as follows: -31.5 dB at 32.4 GHz for AVA-P, -45.5 dB at 19.3 GHz for AVA-LL, -48.2 dB at 16.5 GHz for AVA-LW, and -39 dB at 22.27 GHz for AVA-SL. Because of the enhanced radiated surface area, the reflection coefficient results improved for a wide-bending AVA-LW.

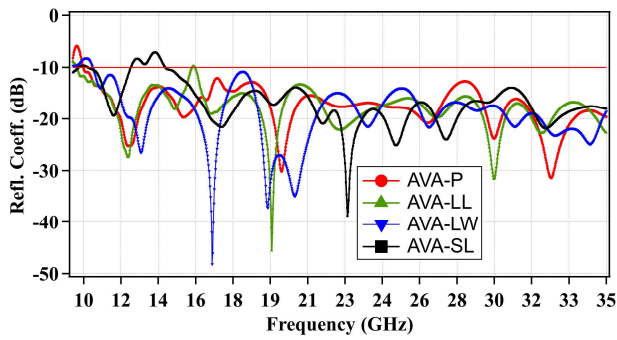


Fig.3. Reflection coefficient comparison graph for various AVA configurations.

B. Gain and radiation efficiency

The simulated gains and radiation efficiency of the four distinct AVA antennas over the whole frequency spectrum are shown in Figs. 4 and 5. The antenna exhibits a gain that peaks at 30 GHz and ranges from 5.2 dB to 12.2 dB. Interestingly, the gain continuously surpasses 10.3 dB in the 10 GHz to 15 GHz frequency band. The other three designs, on the other hand, show gains across the frequency range that are less than 11.7 dB. Notably, when the feed width and flare bending width were equal, the gain increased significantly, as observed in the proposed antenna structure AVA-P.

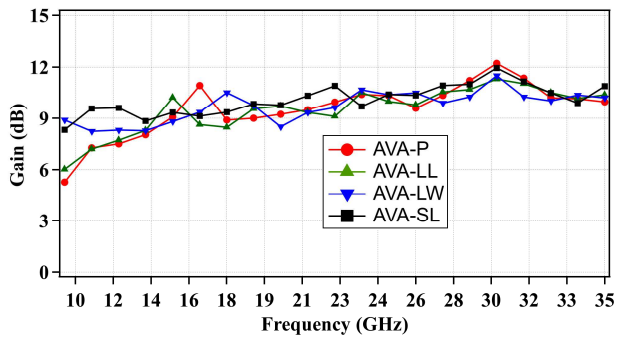


Fig. 4. Gains comparison graph for various AVA settings.

Furthermore, within the obtained band, the simulated radiation efficiency is continuously high, at about 97.7%. The Teflon substrate's minimal loss, as indicated by a loss tangent of 0.0002, is responsible for this remarkable radiation efficiency. Additionally, the efficiency increased greatly for the large leaf structure. As a result, the antenna exhibits exceptional radiation efficiency and gain throughout the whole frequency range.

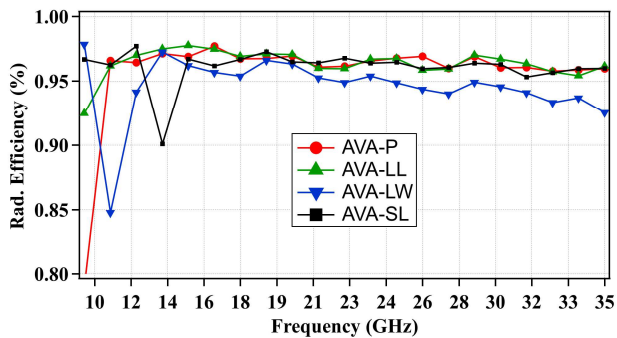


Fig. 5. Radiation efficiency comparison graph for various AVA configurations.

C. Surface current distribution

Understanding and visualizing surface currents is essential in antenna design because they affect key radiation properties, including radiation patterns, impedance, and efficiency. Fig. 6 illustrates the surface current distribution at 24 GHz for various AVA patterns. The distribution of current at lower frequencies tends to focus more on the tapered slots' wider end. The concentration of surface current rises as the leaf size decreases.

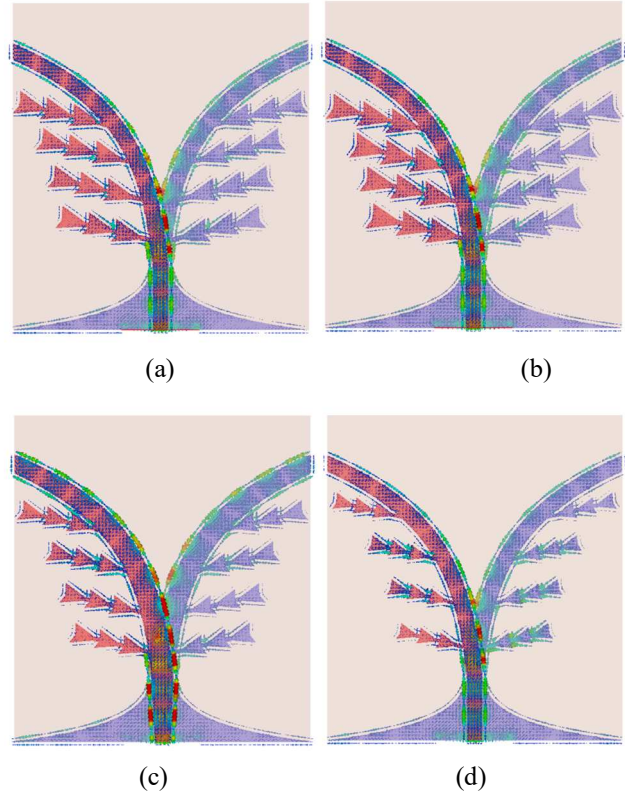
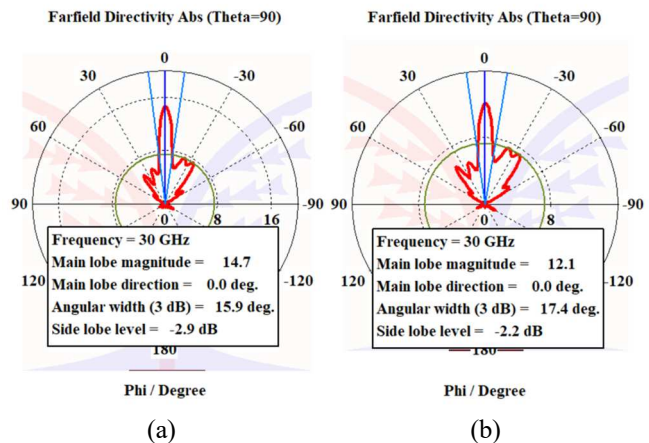


Fig.6. Surface current distribution for all shaped AVA configurations at 24 GHz (a) AVA-P (b) AVA-LL (c) AVA-LW (d) AVA-SL

D. Radiation pattern

A 2D polar plot of the radiation pattern at 30 GHz of the different patterns of the AVA is depicted in Fig. 7. In the high-frequency range at 30 GHz the main lobe direction is 0 degrees. The side lobe level (SLL) of the proposed AVA-P is notably lower than other designs. At 30 GHz, the proposed AVA-P and others demonstrate less tilt compared to other frequencies of AVAs.



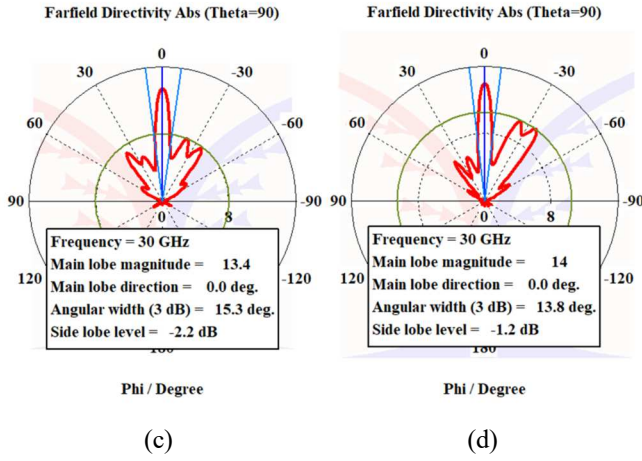


Fig.7. Radiation pattern at 30 GHz for different AVA configurations (azimuthal plane) (a) proposed AVA-P (b) AVA-LL (c) AVA-LW (d) AVA-SL.

Table II represents antenna performance comparison with recently published works, and it shows the improvement that is associated with the proposed work.

TABLE II. PERFORMANCES COMPARED TO OTHER LITERATURE WORKS

Ref. No.	Size (mm ³)	ϵ_r	BW (GHz)	Reflection coefficient (dB)	Gain (dB)
[8]	(60×30×1.6)	4.4	6-18	-22.5	6.7
[10]	(50.8×62×0.8)	4.4	0-20	-34.5	10
[12]	(60×48×1)	4.6	2-14	-40	10
[13]	(160×120×1.57)	2.2	0.8-18	-40	10
[14]	(56×50×0.8)	4.4	1-12	-50	5.2
[15]	(140×66×1.5)	2.94	2-32	-28	9
This work	(54×49×0.8)	2.1	10-35	-31.5	12.2

Here, we can see that in the above works, some antennas have larger sizes, some have limited bandwidth, and others have lower gain and are not in the ultra-wideband (UWB) range. However, in this study, the antenna design is more compact, the frequency range is UWB, and the gain is high.

IV. CONCLUSION

In this paper, different types of leaf structures were proposed for performance enhancement through process optimization. Three techniques were used to compare performance. First, a large leaf structure was designed to improve radiation efficiency. Second, applying a large width of bending in the AVA, resulted in a better reflection coefficient and improved surface current distribution. Third, with a smaller leaf structure, the concentration of surface current increased. Finally, in the proposed AVA (AVA-P), where the feed width (w_1) and flare bending width were equal, the gain increased significantly for this, and the overall results were better than the others. The performance enhancement data for a leaf-shaped antipodal Vivaldi antenna within the 10 GHz to 35 GHz frequency range are as follows: maximum

gain of 12.2 dB at 30 GHz and maximum radiation efficiency of 97.7%. Given its enhanced performance, the proposed antenna is a strong candidate for robust wireless communications in industrial IoT applications.

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